

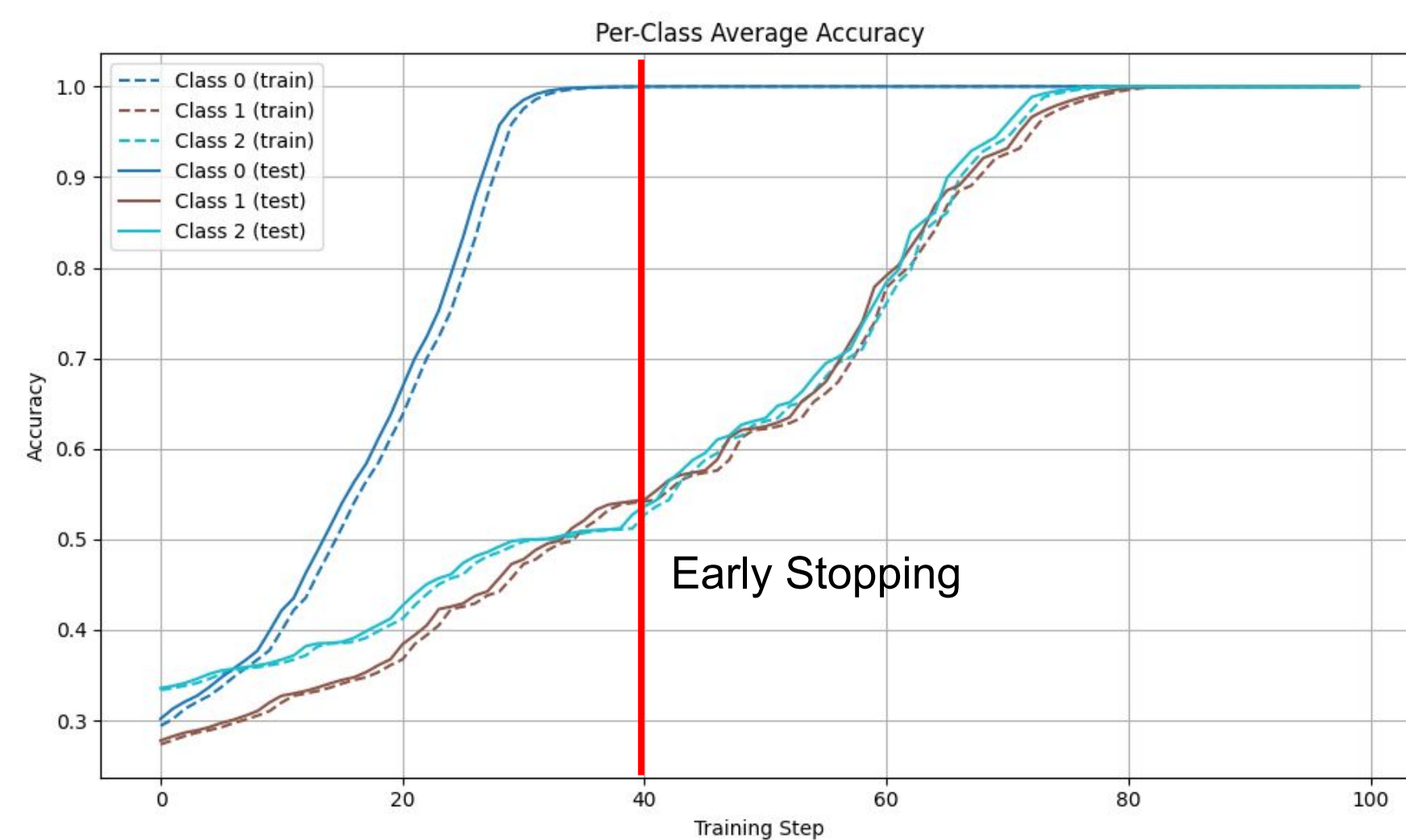
Linear Classification Can Lack Fairness Even With Linearly Separable Data

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Abstract

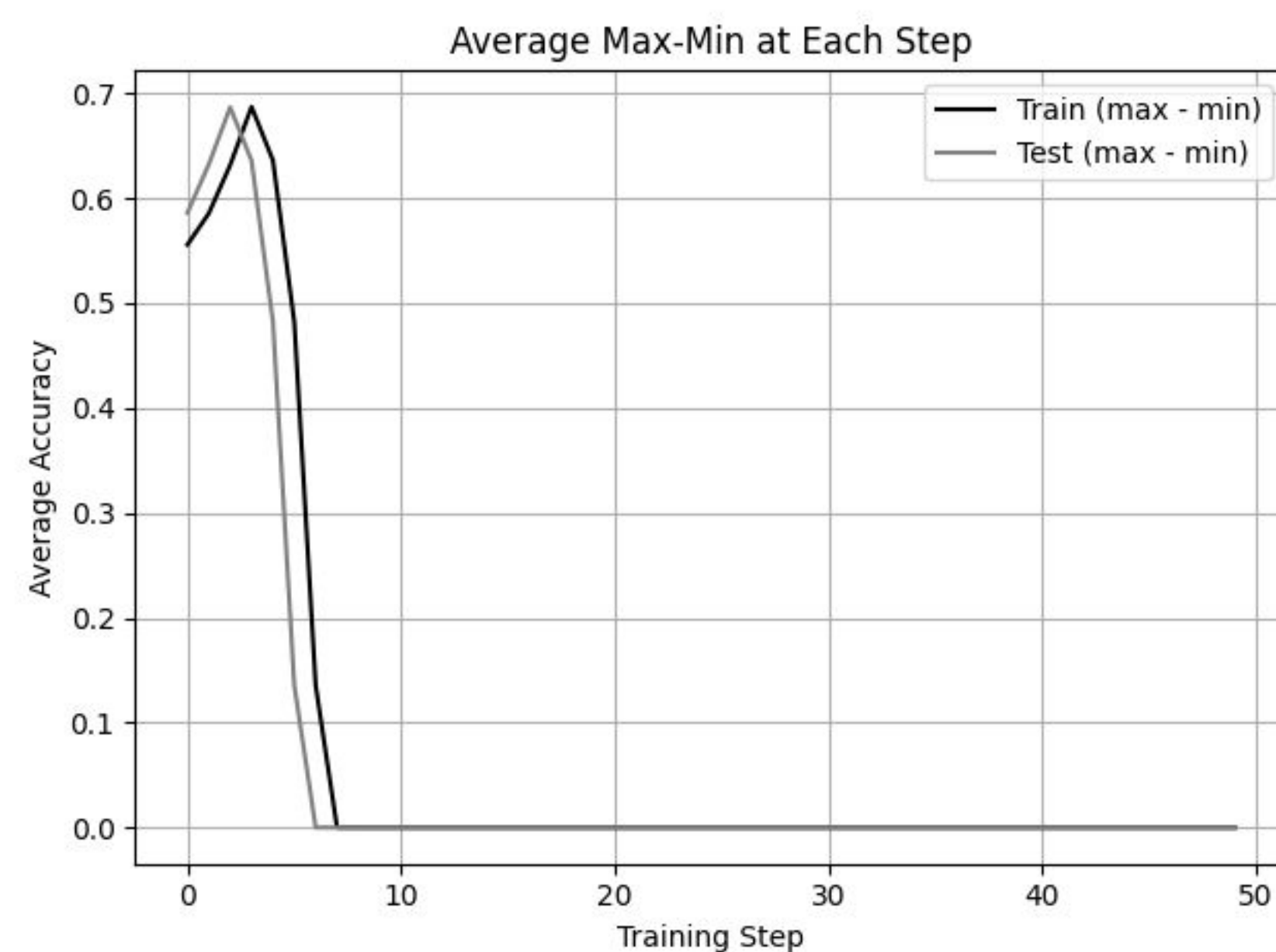
Class bias is a pervasive phenomenon observed across machine learning wherein models learn features that skew their predictions to favor certain classes. The mechanisms that cause these learning imbalances however, remain largely unknown. In this paper we examine per class fairness in the linear regime. We leverage the ideas of Neural Collapse to observe how a convex combination of the input data with points projected onto vertices of an Equiangular Tight Frame(ETF) affects class fairness. We find this convex combination improves similarity of average convergence speed between classes and dive into a preliminary examination of the causes. We also attempt to prove initialization as a cause of class bias.

Convergence Bias



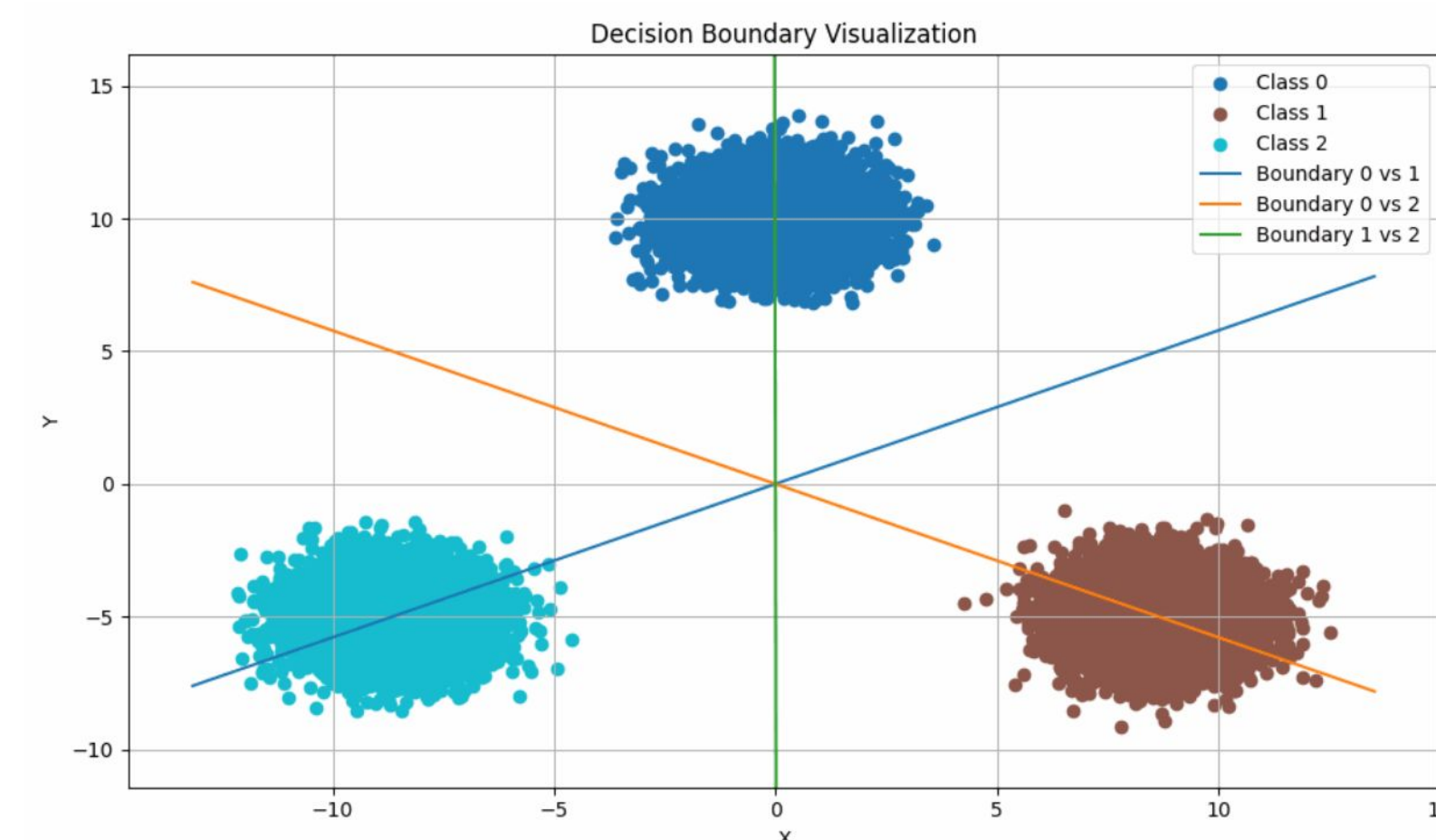
- Demonstrates which classes converge faster on average.
- Regularization techniques like early stopping can cause unfairness in models with differing convergence speeds.

Max-Min Bias



- Maximum Per-Class Classification Accuracy – Minimum at each training step.
- Tells us the average difference between the best and worst performing classes respectively throughout training.

Model Architecture



- Decision boundary between each of the 3 classes
- Line exists where the probability of being in each class is equal

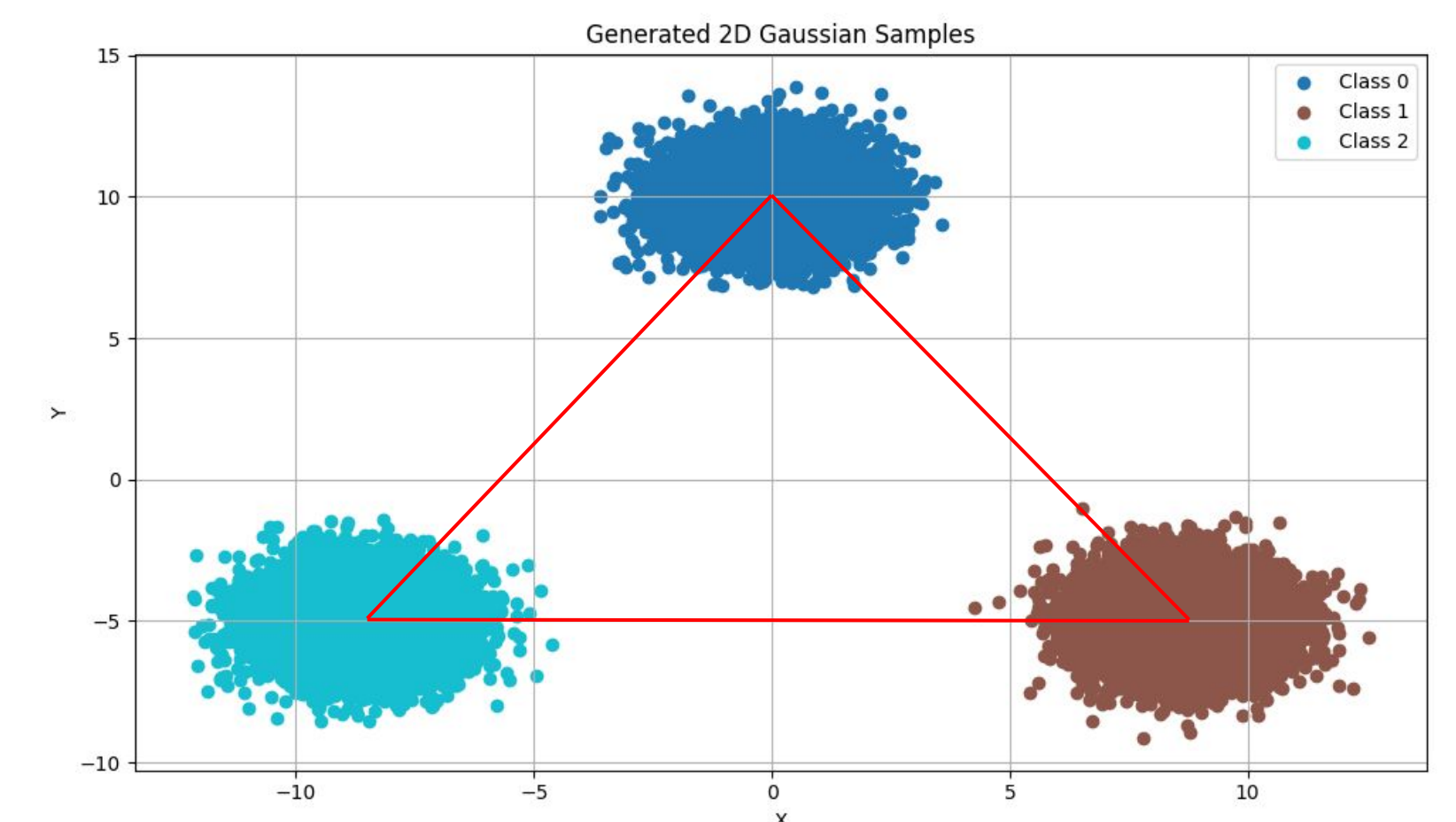
$$Wx + b = y$$

$$\begin{bmatrix} w_{0,0} & w_{0,1} \\ w_{1,0} & w_{1,1} \\ w_{2,0} & w_{2,1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$L = \frac{1}{n} \sum_{i=1}^n (\hat{y}_k - y_k)^2$$

$$(\eta * \nabla L) + W \rightarrow W$$

Projection Methods

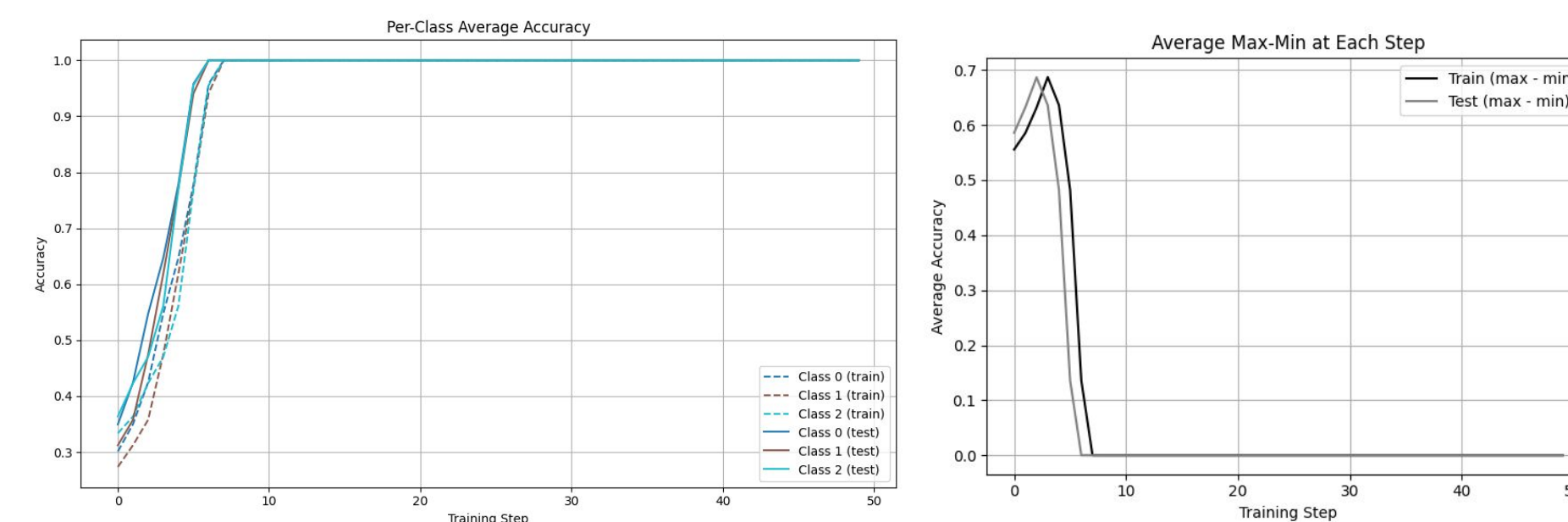


- Scaling & Shifting Methods
- Max, Mean, or Median Norm

$$\hat{x} = (\lambda x) + ((1 - \lambda) * NewPoints)$$

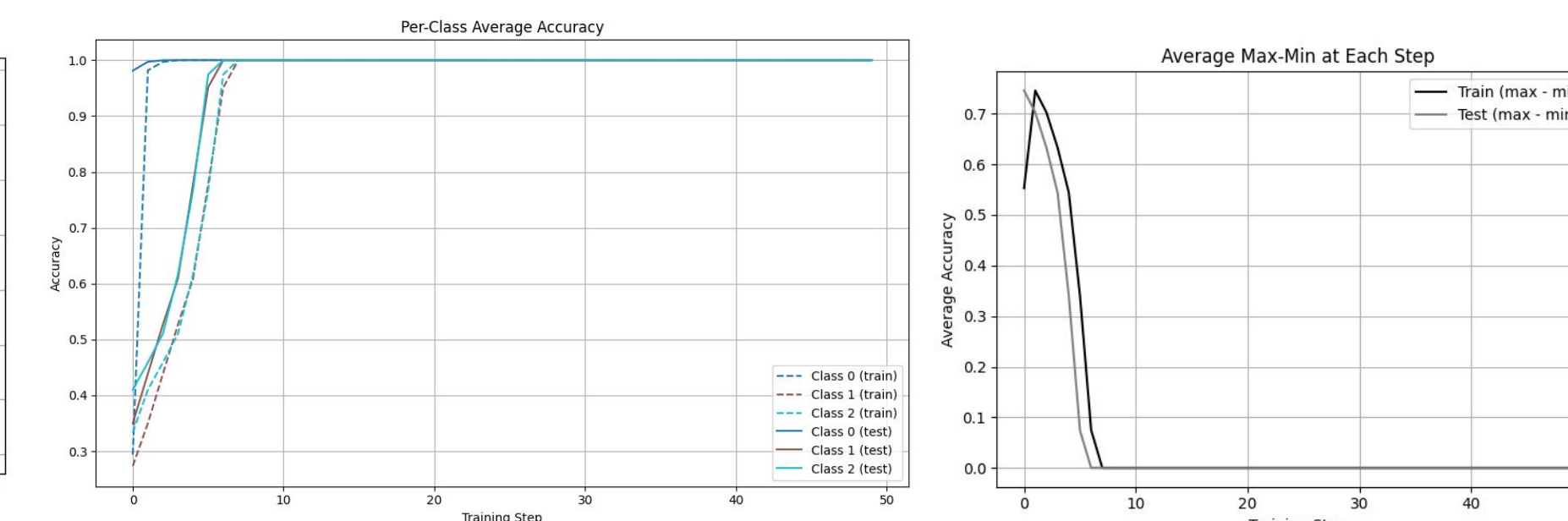
Testing and Results

Equidistant Classes



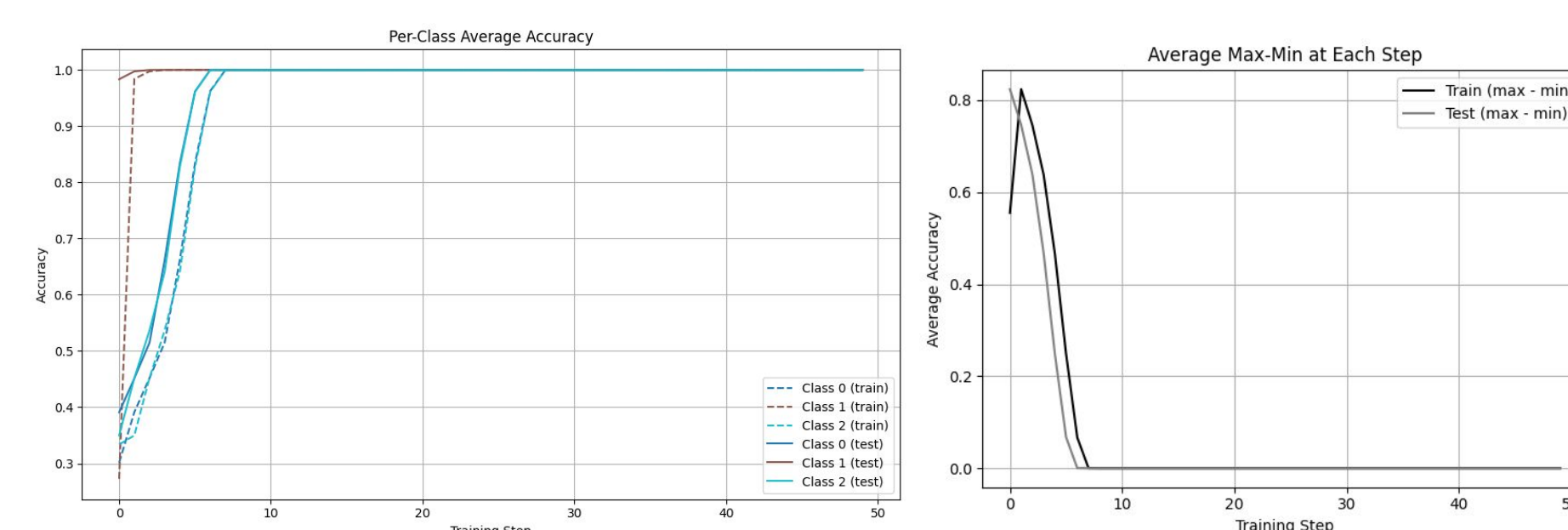
Uniform average convergence speed, but still a large Max-Min. Potentially caused by initialization bias.

Class 0: 2x Farther



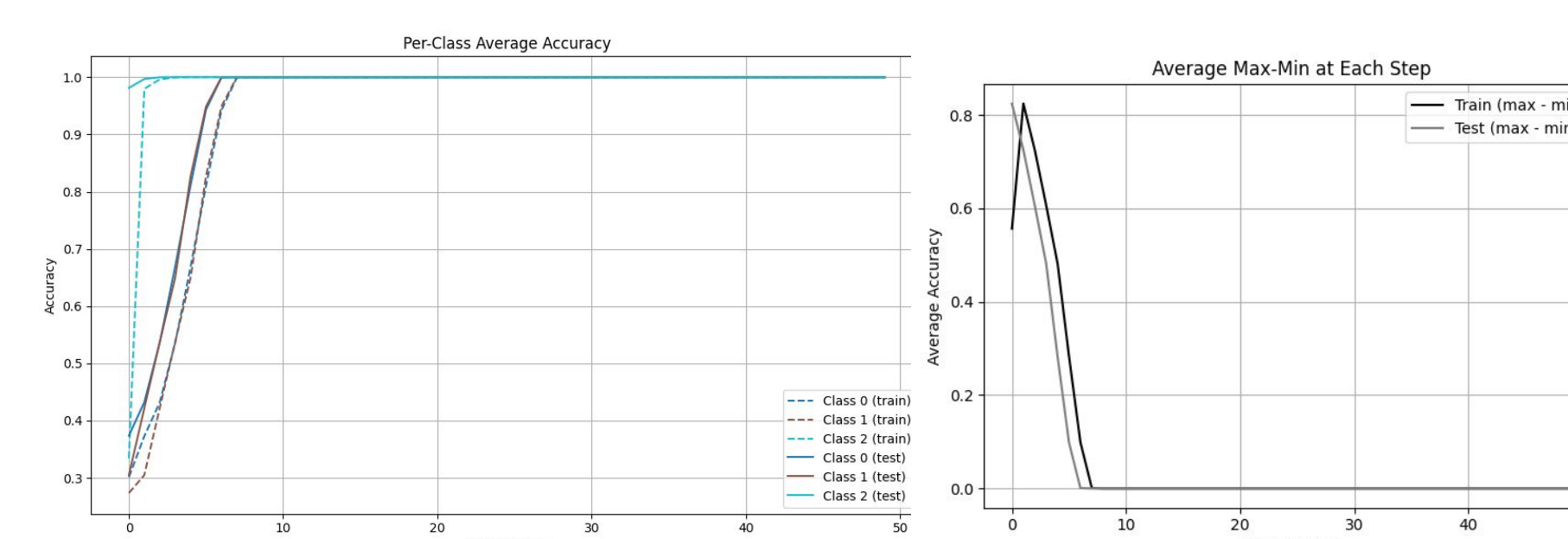
Average convergence speed biased towards the farthest class.

Class 1: 2x Farther



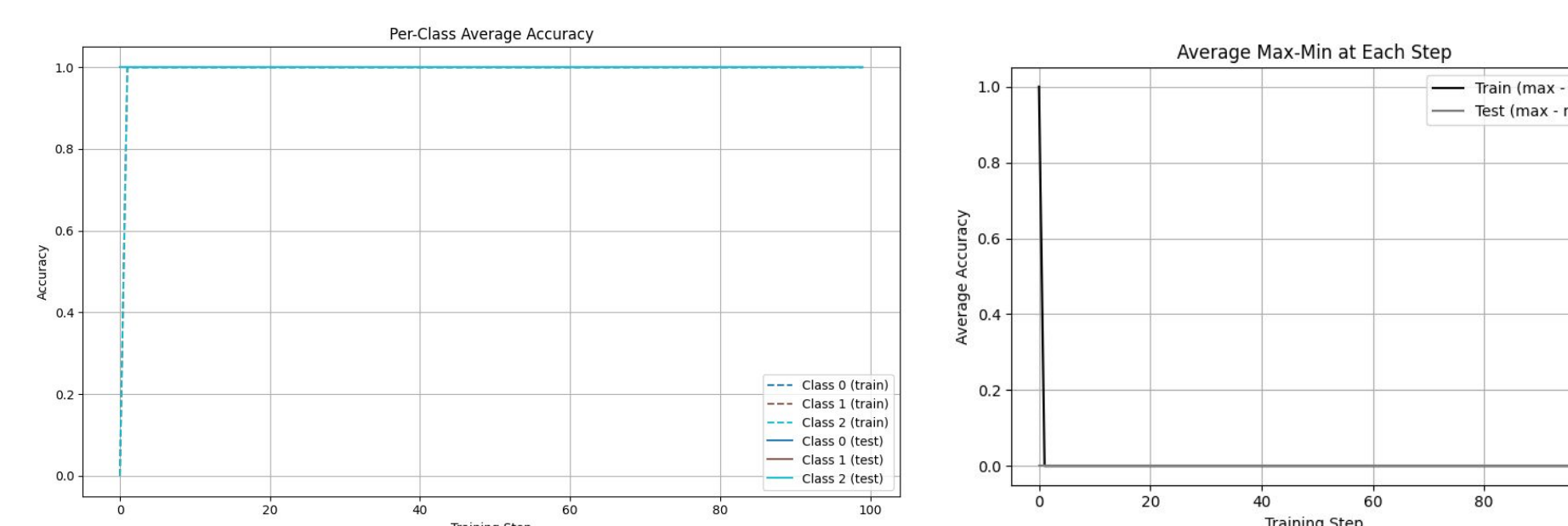
Average convergence speed bias exists regardless of which class is moved.

Class 2: 2x Farther

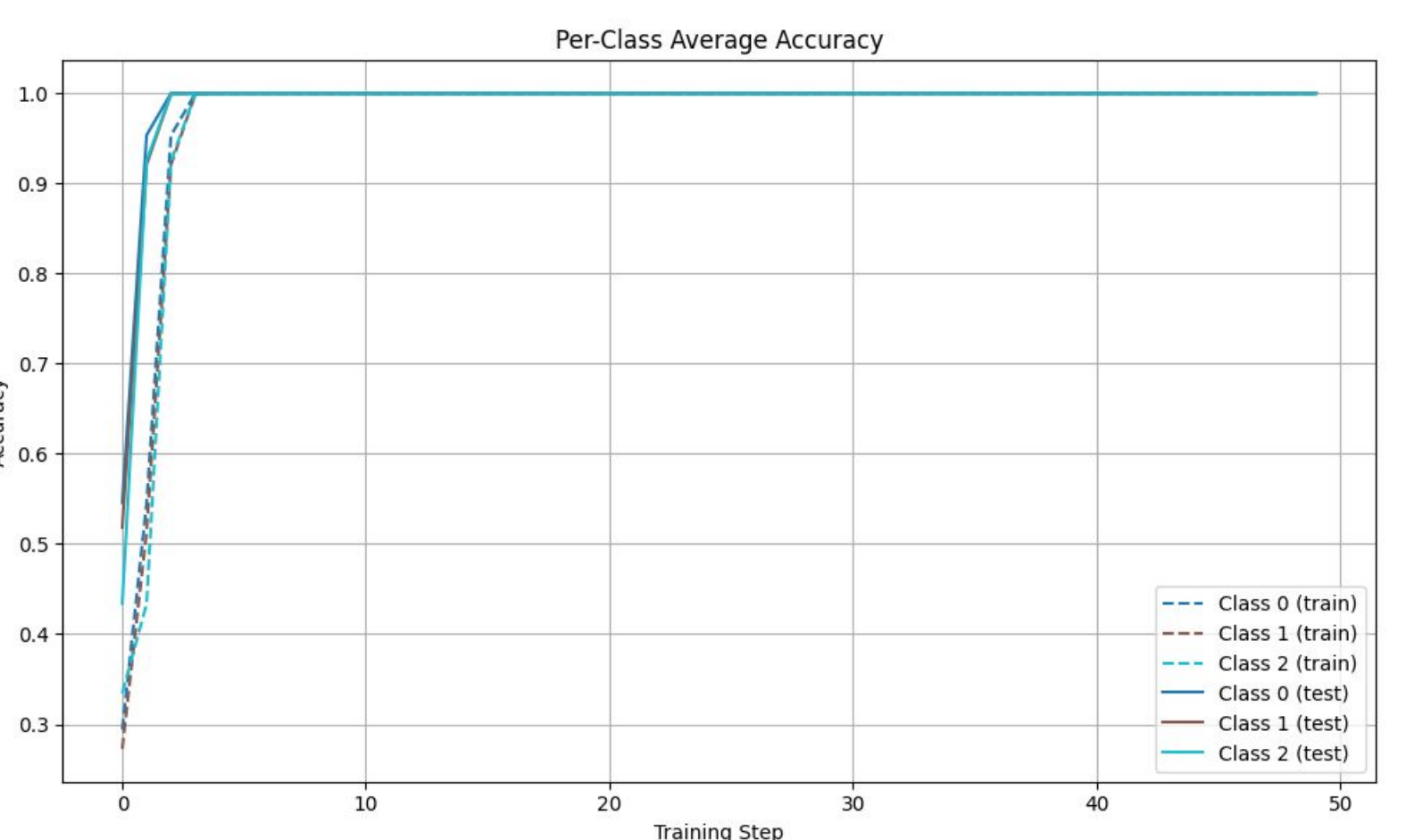


Repeated observation. Could be caused by increased initialization area or convergence properties of SGD.

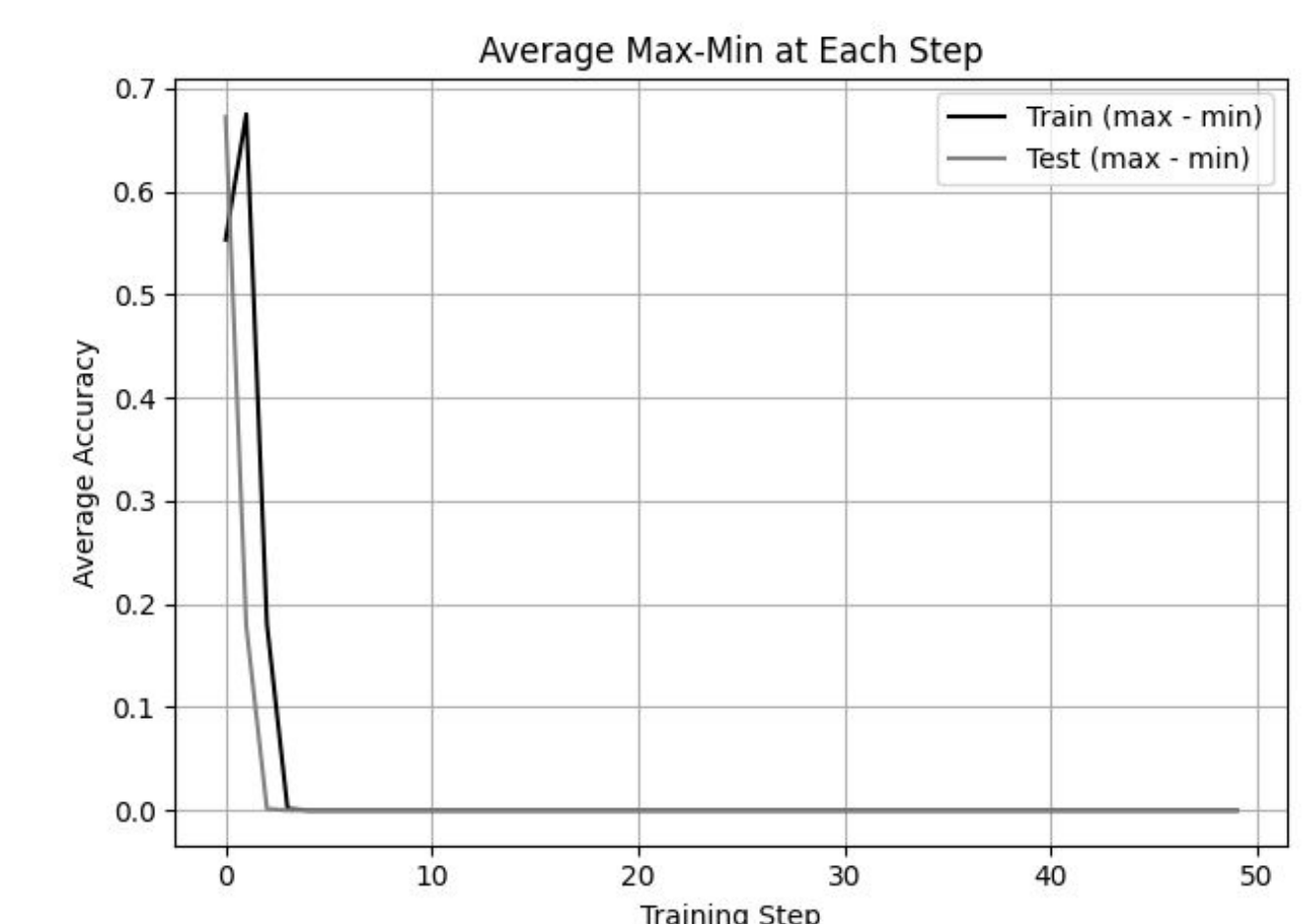
Weights Initialized To 0



Uniform fast convergence with almost instant drop to 0 Max-Min.



- Improved uniformity of average convergence speed



- There still exists a large Max-Min during training

Acknowledgements

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